



RS003 Week 5 — English Worksheet

Topic: Wrap the Ship in a Function · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week you wrap your whole spaceship in a **function** called `draw_ship`. Then you give the function a **parameter** so you can draw the same ship at any X position you like.

Words to know: **function**, **define** (`def`), **call**, **parameter**, **polygon** (nose)

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
def draw_ship():
    pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
    pygame.draw.polygon(screen, WHITE, [(CENTER_X, HEIGHT - 100), (CENTER_X - 20,
HEIGHT - 80), (CENTER_X + 20, HEIGHT - 80)])
```

This defines a function but does not run it yet. What two shapes will it draw when it IS called?


```
screen.fill(SPACE)
draw_ship()
pygame.display.flip()
```

Now the function is called. When does the ship actually appear?


```
def draw_ship(x):  
    pygame.draw.rect(screen, CYAN, (x - 20, HEIGHT - 80, 40, 50))  
  
draw_ship(200)  
draw_ship(400)  
draw_ship(600)
```

The function now takes . How many ships appear, and where?

2 · Run It

 **Type the example, run it, and find the nose**

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))

SPACE = (10, 12, 40)
CYAN = (80, 220, 255)
WHITE = (255, 255, 255)
RED = (230, 60, 60)

def draw_ship():
    pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
    pygame.draw.polygon(screen, WHITE, [(CENTER_X, HEIGHT - 100), (CENTER_X - 20,
HEIGHT - 80), (CENTER_X + 20, HEIGHT - 80)])
    pygame.draw.rect(screen, RED, (CENTER_X - 16, HEIGHT - 30, 8, 10))
    pygame.draw.rect(screen, RED, (CENTER_X + 8, HEIGHT - 30, 8, 10))

clock = pygame.time.Clock()
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    screen.fill(SPACE)
    draw_ship()
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Where does the nose point? What three points does the `polygon` connect?

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The ship function is written but nothing appears on screen.

```
def draw_ship():  
    pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))  
  
screen.fill(SPACE)  
pygame.display.flip()
```

Hint: a function only runs when you **call** it.

Write the fixed code (show the loop body):

Why was it wrong? Why does your fix work?

Bug B — The nose is supposed to point **up** from the body. Right now its tip is below the body so it points down.

```
pygame.draw.polygon(screen, WHITE, [(CENTER_X, HEIGHT - 60), (CENTER_X - 20, HEIGHT  
- 80), (CENTER_X + 20, HEIGHT - 80)])
```

Hint: to point up, the tip point needs a **smaller** Y than the two base points.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should draw three ships at 200, 400, and 600, but Python gives an error about a missing argument.

```
def draw_ship(x):  
    pygame.draw.rect(screen, CYAN, (x - 20, HEIGHT - 80, 40, 50))  
  
draw_ship()
```

Hint: the function needs an x value when you call it.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Give the function a position parameter

Change `draw_ship()` so it takes an `x`, and use `x` everywhere the position appears instead of `CENTER_X`.

Write your new function header line (the `def ...` line):

Draw a whole fleet of ships

Call your function three times with different X values to make a fleet.

Write the three calls you used:

Hint: `draw_ship(200)` draws one ship centred at $X = 200$. Change the number for each one.

5 · Make It 📷

 Build a ship you can place anywhere

Build a program that:

1. has a `draw_ship(x)` function with a body, a nose, and thrusters,
2. is called inside the game loop,
3. draws at least one ship on screen.

Send a **photo or video** of your ship, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I wrapped my ship in a function called ...

I gave it a parameter so that ...

The nose points up because the tip point has ...

To draw more than one ship, I ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video of your ship. Talk through it like you are teaching someone. Try to use these words: **function, define, call, parameter, nose**.

1. Show your ship with its nose.
2. Read your `def draw_ship(x)` line out loud and explain the parameter.
3. Point at where you **call** the function.
4. Change the X value live and show the ship move.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + your photo or video to teacher on KakaoTalk.